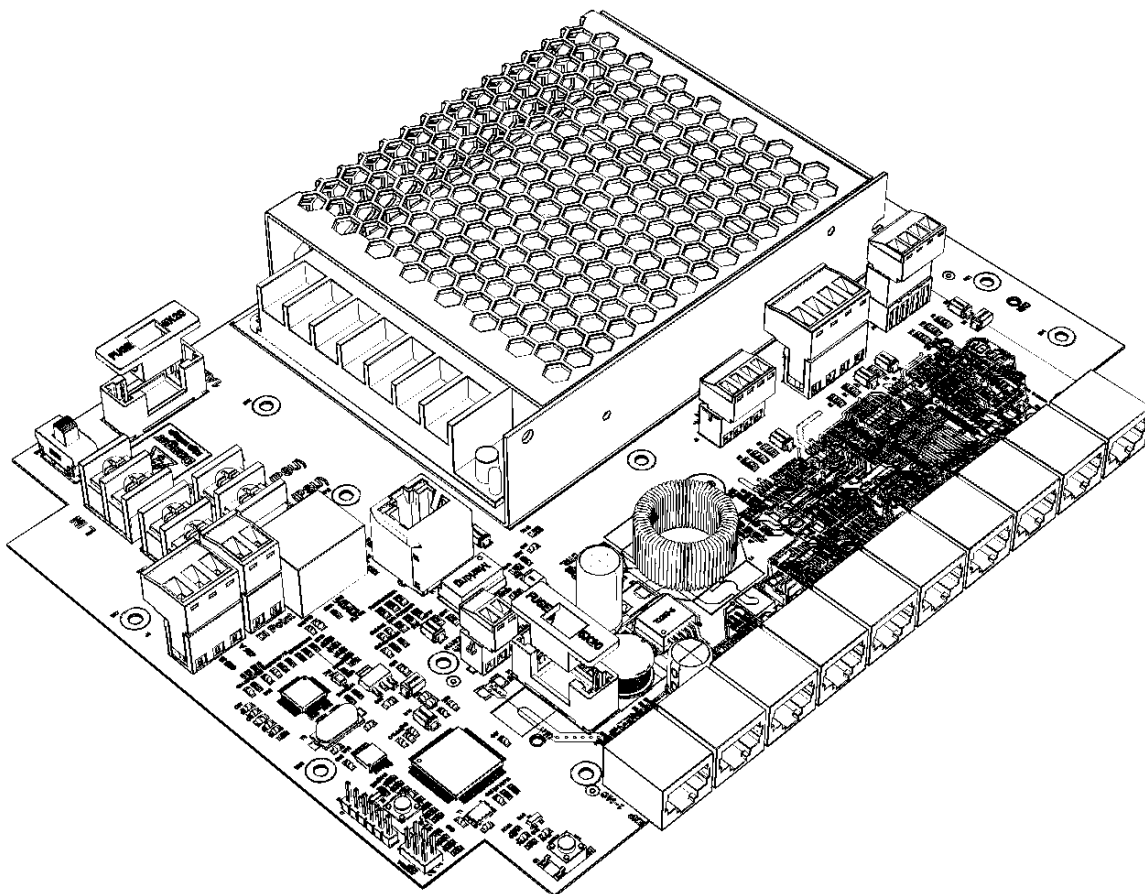




INSTALLATION and USER MANUAL

Defrost on Demand (DoD)

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Thank you for purchasing a City FM's Defrost on Demand Device. So that your device serves you well, please go through this manual and keep it for future use.

Before Using the Device

Important Safety Instructions

- The safety of this equipment complies with industry norms and meets legal equipment safety requirements. However, as manufacturers, we consider it our duty to provide you with the following safety instructions.
- It is essential that you store this user's manual in a safe place so that you can use it in the future.
- Before putting the device into operation for the first time, check that it has not been damaged during transport. Never use a damaged device. If some parts are damaged, contact the supplier.
- The warranty does not cover damages arising from external factors (foods, fire, rodents, etc.), for details please refer to the warranty card.
- Keep packaging materials out of the reach of children!
- Do not place or store the device where it can fall.
- Keep cigarettes and other open flames away from the device. There is a danger of a fire hazard.

General Safety Instructions

- Changing the technical parameters or making any other modifications to the device is dangerous.
- Never disassemble, repair, or alter the unit. Doing so may void the device warranty and may cause fire, electric shock, over-heating, or short-circuiting.
- Do not store or place & flammable substances near the device.
- Never use accessories, which have not been supplied with this device or which have not been explicitly recommended by the manufacturer.
- Exercise caution when maintaining your equipment, be sure all components are firmly screwed or latched into position.
- Before connecting this device to a power socket, make sure the voltage stated on the label of the device corresponds to the voltage in your wall socket.
- Ensure that the power plug contacts do not come into contact with water or moisture.
- Do not place the device in the vicinity of an open flame or devices that are sources of heat.
- Do not place the power cord near hot places or over sharp objects. Do not place heavy items on the power cord. Position the power cord so that it will not be stepped on or tripped over.
- Disconnect the device from the power socket after every use.

- Do not disconnect the device from the power socket by pulling on the power cord. This could damage the power cord or the power socket. Disconnect the cord from the power socket by gently pulling the plug of the power cord.
- If the power cord is damaged, have it replaced at a specialized service center. It is forbidden to use the device if it has a damaged power cord or plug.
- To avoid the danger of injury by electrical shock, do not repair the device yourself or make any adjustments to it. Have all repairs or adjustments of the device performed at an authorized service center.
- Tampering with the device during the warranty period may void the warranty.

Important safety measures

Carefully read all the important safety instructions. Otherwise, there exists a risk of an accident or a serious injury.

Risk of death/injury by electrical shock

  *High voltage present inside the device. Risk of death or severe injury if proper safety measures are not adopted.*

- High voltage is present inside the device. Don't touch any component with bare hands. There is a risk of severe electric shock.
- Do not plug or unplug the power plug with wet hands. Doing so may cause electrical shock.
- If the device is damaged, do not touch the damaged section.
- Do not place in or drop into water or other liquid.
- Do not disassemble the device.
- Do not get any water or cleaning solution on the top of unit or power plug.
- Do not use a loose or faulty electrical outlet. Doing so may cause fire or short circuit.
- Do not use any outlet or wiring equipment beyond its specified rating.
- Never touch the device with wet hands or feet.
- Never use a damaged power cord as it may result in a fire, electrical shock, overheating or short circuiting.

Risk of fire or Burns

  *This is an electrical device and is susceptible to fires.*

- The enclosed parts become hot during operation. Avoid touching any part, there is a risk of burns.

- Do not store or place any flammable substances near the device.
- Keep cigarettes and other open flames away from the device. There is a danger of a fire hazard.
- Do not expose the device to temperatures in excess of 50 °C.
- Do not expose the device to direct sunlight.

Risk of Explosion

 *Electric components may explode if not disposed properly.*

- Do not subject the device to strong impacts.
- Do not crush or puncture any component.
- Do not dispose of the device in a fire.

Risk of damage to device

 *This is an electrical device. Don't expose the device to high pressures or impacts.*

- Do not allow the device to get in contact with liquids.
- Do not subject the device to high pressures.
- Do not place any objects on top of the device pack.

Risk of Radiations or High Frequencies

 *Extended close distance exposure to this electrical device must be controlled.*

- To avoid damage to health, avoid close distance (7.87in/20cm or less) to device for a long time.
- All electrical installations must be carried out in accordance with the local electrical standards.

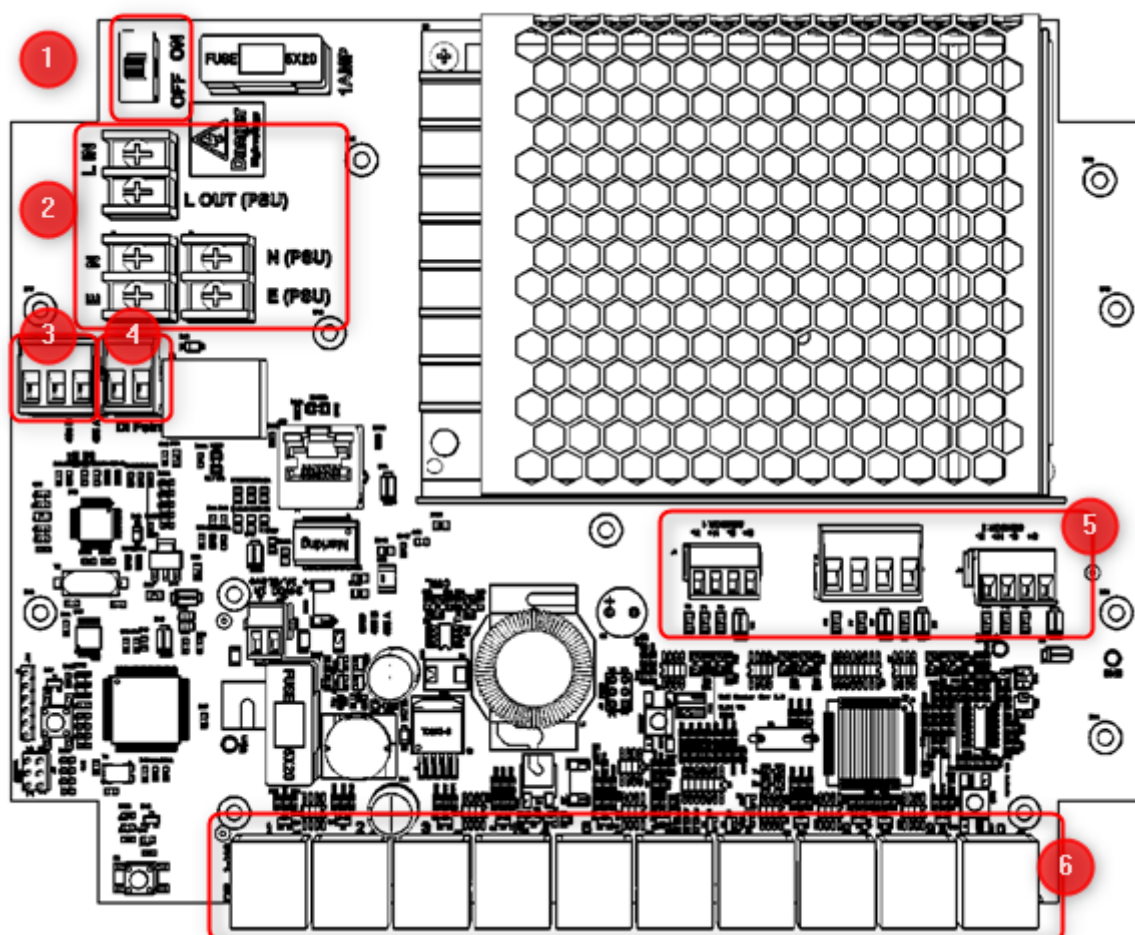
Child safety

 *Don't allow children to play with the device. It has electrical components inside.*

- Do not allow children to play with the device. The device has electric components inside.
- This device is not a toy and should never be given to children without strict and close supervision of an adult.
- Do not leave children near the device without supervision.
- Keep packaging materials out of the reach of children!

Description of the Device

Device Description



Sr. #	Part Name
1.	ON / OFF Switch for 24VDC Transformer
2.	220 VAC Supply from the Freezer Controller
3.	24 VDC Relay Module to freezer's controller DI point
4.	Terminal for Modbus cable
5.	Terminal for each temperature sensor strip
6.	To each DoD Slave with Cat5e Cables

Note: The final device may differ from the description.

Device's Key Characteristics

- Efficient power utilization: The "Defrost on Demand" device optimizes energy consumption in industrial refrigerators. By automatically initiating defrost cycles, when necessary, it minimizes power usage, resulting in significant energy savings and reduced electricity costs.
- Consistent and reliable airflow: With continuous airflow monitoring every 15 minutes, our device ensures a consistent and reliable cooling performance. By automatically initiating defrost cycles when reduced airflow is detected, it prevents ice buildup on the evaporator coils, maintaining efficient airflow and preserving optimal temperatures for stored goods.
- Compatibility with various industrial refrigerators: Designed for versatility, the device can be easily installed in different models and makes of industrial refrigerators. It seamlessly integrates with existing refrigeration systems, making it a convenient and adaptable solution for various setups.
- Prolonged equipment lifespan: By preventing ice formation on the evaporator coils, the device extends the lifespan of industrial refrigerators. It minimizes wear and tear caused by ice buildup, reducing the frequency of breakdowns and the need for costly repairs. This leads to improved reliability and longevity of the equipment.
- Simplified maintenance routines: The automated defrosting process eliminates the need for manual monitoring and intervention, streamlining maintenance operations. Maintenance personnel can focus on other critical tasks instead of constantly checking and initiating defrost cycles. This improves efficiency and productivity in the maintenance workflow.
- Environmental consciousness: "Defrost on Demand" promotes sustainability by optimizing defrost cycles, reducing energy waste, and minimizing environmental impact. It aligns with eco-friendly practices and contributes to a greener future.
- Quick and easy installation: Installing the device is a straightforward process that requires minimal modifications to the existing refrigeration system. Once installed, it immediately begins delivering the benefits of automated defrosting, improved energy efficiency, and consistent performance.

Device Assembly & Installation

	The device must be installed by a qualified technician only. Improper installation may lead to issues resulting in damage to individuals or device.
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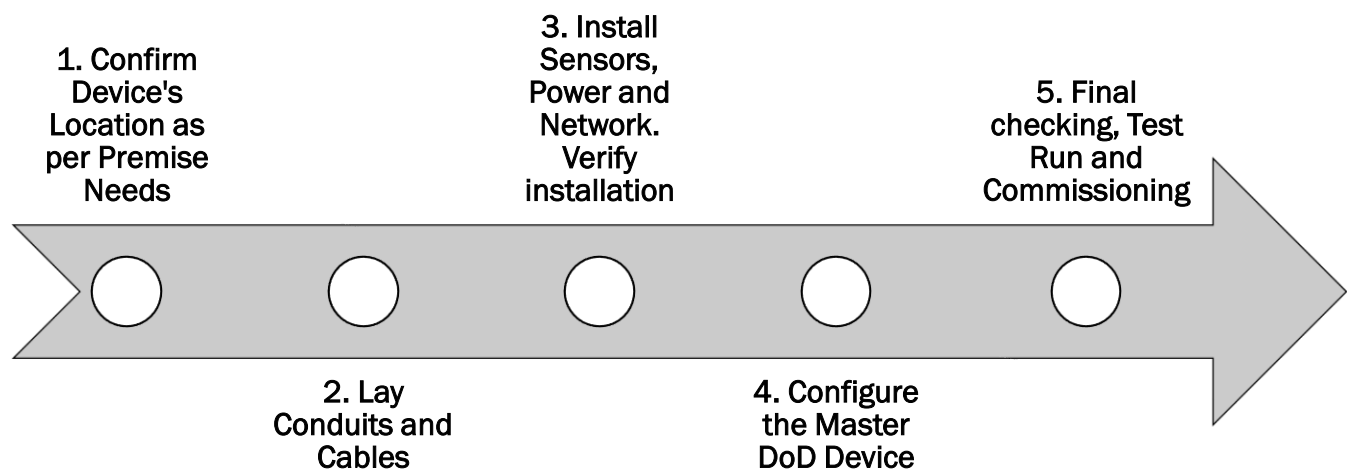
Assembly Prerequisites

Please make sure that the following pre-requisites are met before installation.

1. Adequate power wiring is in place.
2. RS485 wiring is in place and functional.
3. Temperature sensing wiring is in place and functional.
4. There are no flammables or explosives stored nearby.
5. There is adequate air passage in the area.
6. There is no excessive dust and dirt in the installation area.

Typical Installation Workflow

Although each installation is tailored to the specific needs of the premise, all installations follow the following core steps.



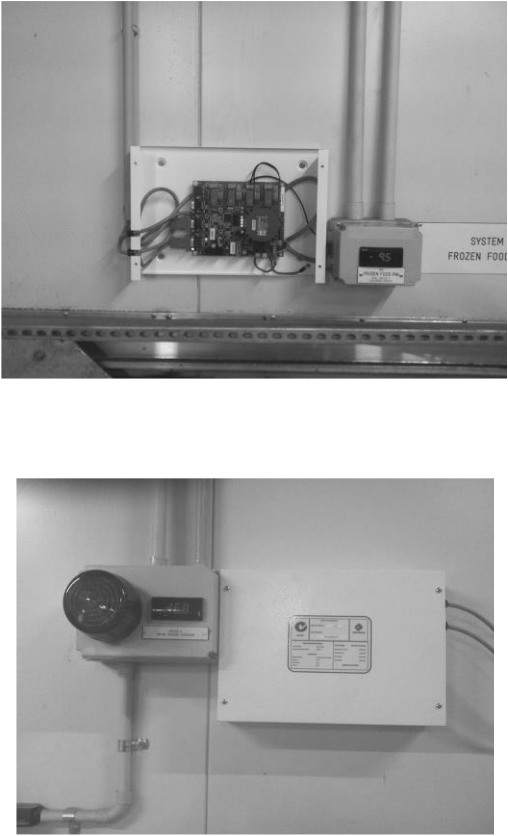
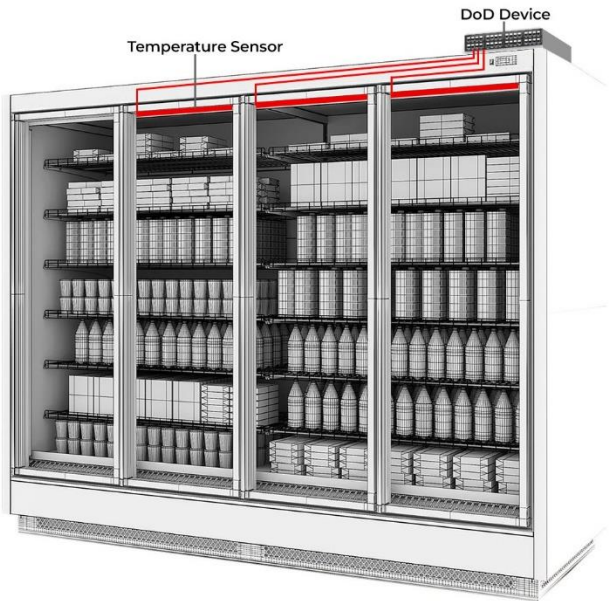
Core steps in any typical DoD installation

Step by Step Device Assembly & Installation

Follow these simple steps to accurately assemble and install the DoD device.

Step 1. Mount the DoD Unit.

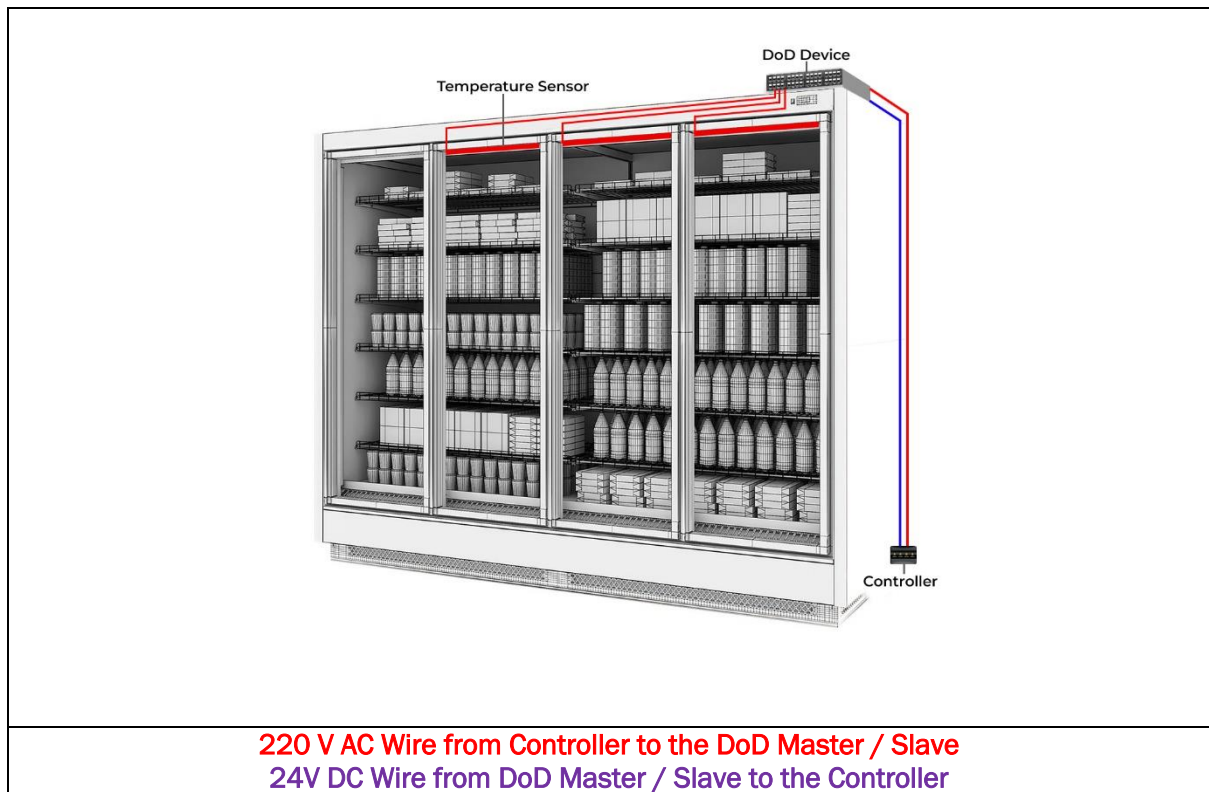
- a. For Freezer Rooms, mount the unit on the front on each room for easy access.
- b. For typical industrial freezers, install DoD on top of the freezer. For reference see figures below.

	
DoD Unit Mounted in a room <i>Typical for Freezer Room</i>	DoD Unit on Freezer Top <i>Typical for Individual Freezers</i>

	Note that the DoD Unit must have power all the times.
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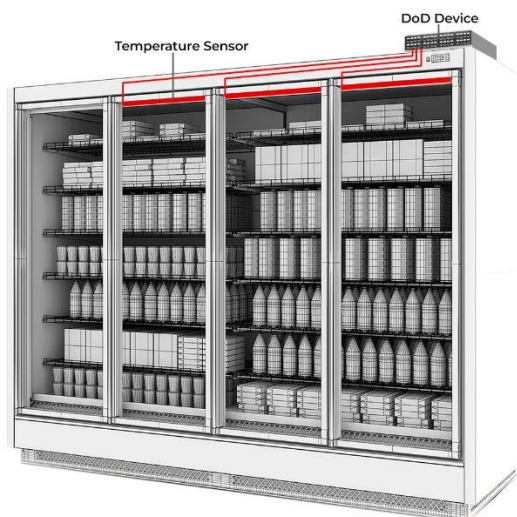
Step 2. Connect Input and Output Power lines.

- a. Pass the wires through the conduit and establish a secure connection of the 220V AC from the controller to the DoD Master/Slave.
- b. Now pass the 24V DC relay output wire from DoD Master/Slave and securely connect to the Controller DI point with conduit.



Step 3. Install the temperature sensor (*for individual freezer setup*).

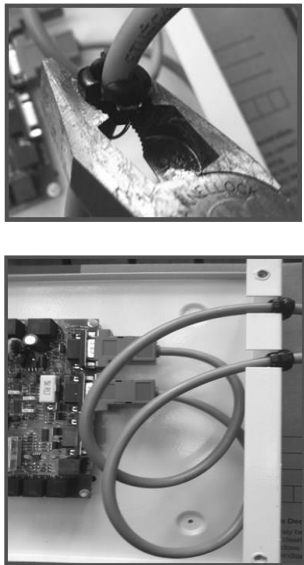
- Place the temperature sensor strip behind the supply louvre.
- Ensure that the sensor strip is affixed uniformly along the louvre.
- Install one strip per freezer.
- See the figure below for reference.



Step 4. Install the air-flow sensor (*for freezer room setup*).

- Mount the air-flow sensor behind the evaporator facing in the same direction as the evaporator fins. Sensors are to be mounted behind the evaporator, equal distance between the rear face of the evaporator and the back of the room.

- b. Mount directly behind each fan for 2 x fan evaporators. For longer evaporators with more than 2 fans, measure the face of the coil and mount the sensors 1/3 in from each end.
- c. Alternatively, the sensors can be mounted through the ceiling of the freezer room or sensors can be mounted on brackets inside the room where access is limited.
- d. See the figure below for reference.

<i>Example Illustration: Typical Through the ceiling installation</i>		
		
<i>Silicon vapor seal around the sensor sleeve</i>	<i>Internal Mount installation using additional bracket</i>	<i>Typical Strain relief gland installation</i>

	Additional brackets if needed for specific installation are not supplied.
	2 Airflow sensors (600mm or 300mm) in a shipping tube are supplied with the package.
	Before applying the strain relief glands, pinch the gland first.
	Proper installation of the temperature sensor is critical for correct operation of the equipment.

Step 5. Lay Cat-5 cables and connect appropriately.

- a. As per the needs of the premise, lay cat-5 cables of required lengths.
- b. Build appropriate cable termination points and connect each cable securely.
- c. Test that the connections are correct.

Device Operation

	Please follow these steps for safe and efficient operation.
	Please do not try to alter any wiring connections or technical parameters as this may result in permanent damage to device or may cause an explosion.

Configure the Controller

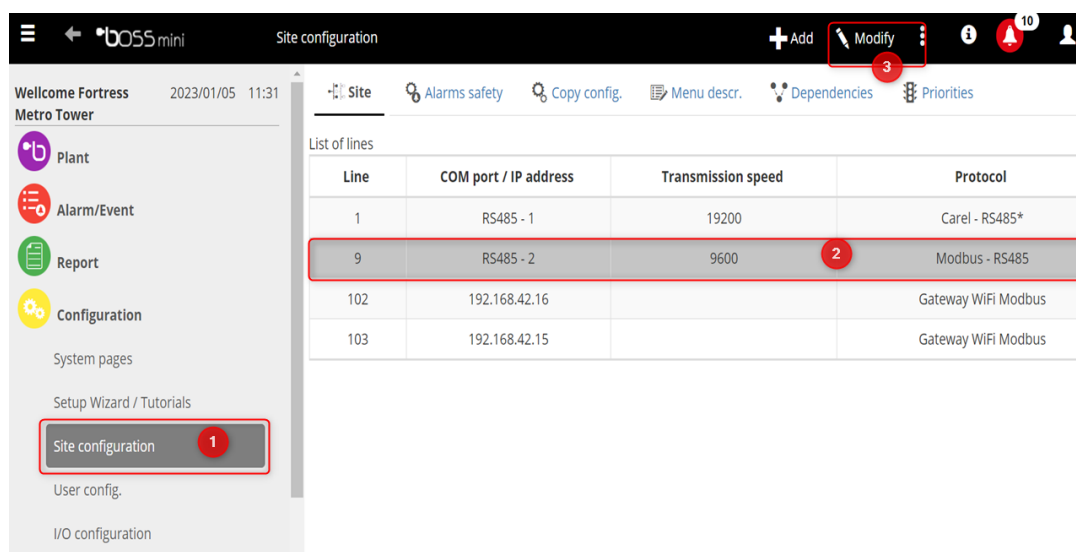
Assign Master Modbus ID

1. Connect Cat5e RJ45 (Port 1-8 any) to PC via serial Port
 - a. Pin4 Serial Port = 485 +
 - b. Pin5 Serial Port = 485 -
2. Open Putty or similar serial command software and enter the command **MSA 10** to assign Modbus ID 10 to the controller.
 - a. To display the Modbus address, type the command **MSA**.
3. Press and hold the reset button.

Add New Device in Carel Boss Application

	To launch the Carel Boss App, type 192.168.42.1 and login using your credentials.
	Please verify that you have connected Cat5e RJ45 cable to Boss system via serial RS485 (Pin4 = 485 + & Pin 5 = 485 -)

1. Navigate to **Site Configurations**, then select the Modbus – RS485 and click **Modify**. Refer to the screen below for details.



2. Enter the **device name** and select the address start option and click the **“+”** button to increase or click **“-”** to decrease the number. Refer to the screen below for details.

Site configuration - Edit

Connection

Protocol: Modbus - RS485, COM port: RS485 - 2, Baud rate: 9600, Poll Delay (ms): 50

Devices

Device name: City FM DoD (1)

From address: 1 (2), To address: 1 (3)

Serial address	Device model	Device description	Disable	Config.	Settings	Import
10	pRack pR300 Modbus Line 1	pRack pR300 Modbus Line 1 - LT				
11	pRack pR300 Modbus Line 2	pRack pR300 Modbus Line 2 - MT				
17	City FM DoD	City FM DoD - 17				

Parameter Configurations

- Once the device is added, navigate to **Device > Parameters** to view the screen as shown below.

Devices / device detail

Parameters

Devices: City FM DoD - 10 (16)

Value	New	U.M.	Short desc	Description
120			Cabinet ID - Master	Cabinet ID - Master
600			SensorModel - Master	SensorModel - Master
14042		ohm	Chan 1 Zero Deg R - Master	Chan 1 Zero Deg R - Master
1684			Chan 1 Temp CoEff - Master	Chan 1 Temp CoEff - Master

- Enter appropriate values for the highlighted attributes. The default values are already set.
- Scroll down to locate the parameters and their values shown in the screen below.

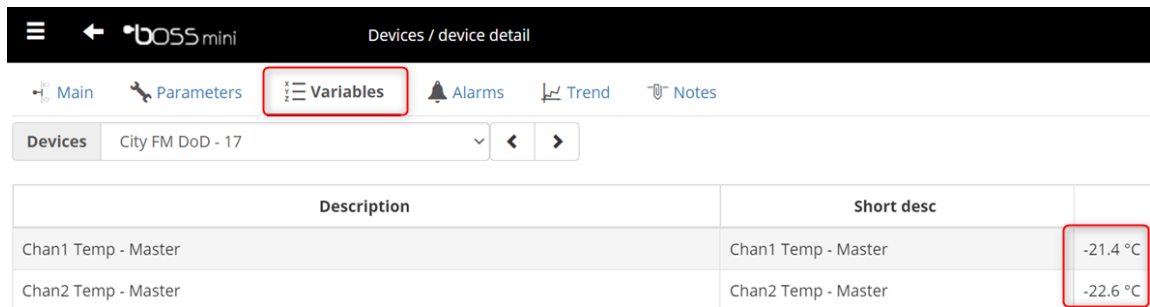
Devices / device detail

Parameters

Devices: City FM DoD - 17

Value	New	U.M.	Short desc	Description
4	24		Airfail - Master	Airfail - Master
0	0		Airtest - Master	Airtest - Master
24	24	hour	Definterval - Master	Definterval - Master
8	6	hour	DefSep - Master	DefSep - Master
173	50	min	Deftime - Master	Deftime - Master
900	900	min	Spdtst - Master	Spdtst - Master

- Once the parameter values are edited, press the **SET** button.
- To view channel temperatures, navigate to **Device > Variables** to view the screen shown below.

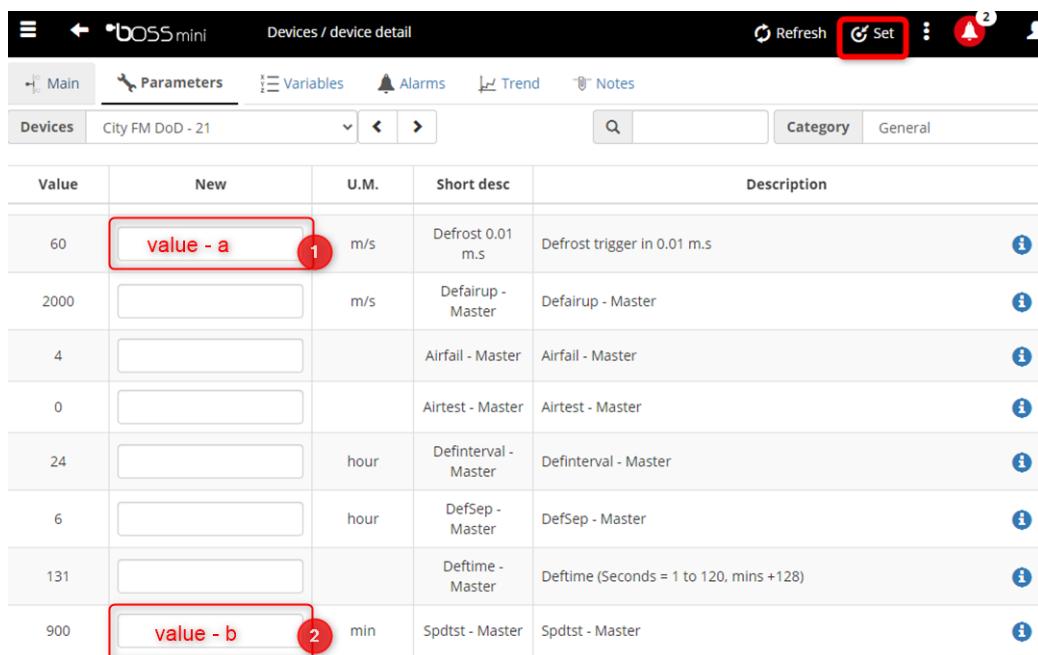


Description	Short desc	
Chan1 Temp - Master	Chan1 Temp - Master	-21.4 °C
Chan2 Temp - Master	Chan2 Temp - Master	-22.6 °C

Test Airflow

Follow these simple steps to check and test airflow.

1. Navigate to **Device > Parameters** to view the screen as shown below.



Value	New	U.M.	Short desc	Description
60	value - a	m/s	Defrost 0.01 m.s	Defrost trigger in 0.01 m.s
2000		m/s	Defairup - Master	Defairup - Master
4			Airfall - Master	Airfall - Master
0			Airtest - Master	Airtest - Master
24		hour	Definterval - Master	Definterval - Master
6		hour	DefSep - Master	DefSep - Master
131			Deftime - Master	Deftime (Seconds = 1 to 120, mins +128)
900	value - b	min	Spdtst - Master	Spdtst - Master

2. Change the **value-b** (ref. screen above) value to “300” to enable airflow testing every 5 minutes.
3. In the “**Variables**” tab, check the Airflow values. Also notes values in the “**Trend**” tab.
 - a. For example, assume that the noted airflow value is **0.9 m/s**
4. Now, calculate the Defrost trigger value by this formula and set the value in **value-a**.
 - a. Formula: $(\text{NotedValue} \times 100) - 30$
 - b. Example: $(0.9 \times 100) - 30 = 87$
5. Now reset the **value-b** to **900**.
6. Click the **SET** button.

Drain Time Setting

By default, the drain setting is set to **2 minutes**. If needed this time can be changed to a maximum of **5 minutes**.

- To change drain timing, type the **drain** command
 - **drain 5** will set the drain time to 5 minutes.

Sensor Model Setting

The sensor model setting configures the airspeed sensor setting. This setting must match the length of sensors installed in each room i.e., 300mm or 600mm.

- To change sensor model setting, type the **sensormodel** command
 - **sensormodel 600** will set the sensor to 600mm model.
 - **sensormodel 300** will set the sensor to 300mm model.

Defrost Time Setting

The defrost time setting configures the length of the defrost cycle.

- To change defrost time setting, type the **deftime** command
 - **Deftime 15** will set the defrost cycle length to 15 seconds.

Important Commands



Do not change any parameters using these commands unless you understand the impacts.

Turn Data-stream on / Off

As a default setting, the DoD controller refreshes the data on the screen every 3 seconds. This continuous information flow is called Data-stream.

- To turn data-stream On, type the **data** command
 - **Data 0** will turn off the data-stream
 - **Data 1** will turn on the data-stream

AirTest

The airstest command runs airspeed tests and displays the results.

- To run airspeed tests, type the **airstest** command
 - **airstest 1** will run airspeed tests using sensor-1.
 - **airstest 2** will run airspeed tests using sensor-2.

Basic Troubleshooting Guide

	This quick reference is not a repair or maintenance guide. Efforts to alter device parameters may result in permanent damage to the device.
	The electrical parts inside the device are non-serviceable, do not try to modify them in any way. It may result in electric shock or serious injury.

The DoD master relies input from the sensors to operate properly. It is important that the sensors are regularly checked for any excessive buildup that may hinder the performance of the sensor.

No Airflow

Check for the possible reasons mentioned below;

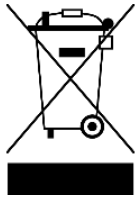
- Check if the DoD Master has power. If not, restore the power.
- Check if the RJ45 cables connections are connected securely. If not, unplug the cables and reconnect.
- Using the cable tester, validate if any of the RJ45 cables is broken. If found, please replace the cable, reconnect and recheck.
- Check if there is a buildup of scale / excessive dirt or ice at the airflow sensor. Remove the scale / excessive dirt or ice and recheck.
- Check if the communication between DoD master and slave is faulty. If found, press the reset button on the master and recheck.

Technical Specifications

Technical Specifications

Specifications	Values

Disposal of Used Packaging Material



The meaning of the symbol on the device, its accessory or packaging indicates that this device shall not be treated as household waste. Please, dispose of this device at your applicable collection point for the recycling of electrical & electronic equipment waste. Alternatively, in some states of the European Union or other European states you may return your devices to your local retailer when buying an equivalent new device. The correct disposal of this device will help save valuable natural resources and help in preventing the potential negative impact on the environment and human health, which could be caused as a result of improper liquidation of waste. Please ask your local authorities or the nearest waste collection center for further details. The improper disposal of this type of waste may fall subject to national regulations for fines.

For business entities in the US

If you wish to dispose of an electrical or electronic device, request the necessary information from your seller or supplier.

Changes to the text, design and technical specifications may occur without prior notice and we reserve the right to make these changes.